

Binary Search

Binary search. Given `value` and sorted array `a[]`, find index `i` such that `a[i] = value`, or report that no such index exists.

Invariant. Algorithm maintains $a[\text{lo}] \leq \text{value} \leq a[\text{hi}]$.

Ex. Binary search for 33.

[illegible]

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6	13	14	25	33	43	51	53	64	72	84	93	95	96	97
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
↑							↑							↑
lo							mid							hi

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Diagram illustrating the initial state of the array and pointers for the binary search algorithm:

Index	Value
0	6
1	13
2	14
3	25
4	33
5	43
6	51
7	53
8	64
9	72
10	84
11	93
12	95
13	96
14	97

Pointers and their corresponding indices:

- lo** (low) points to index 0.
- mid** (middle) points to index 3.
- hi** (high) points to index 6.

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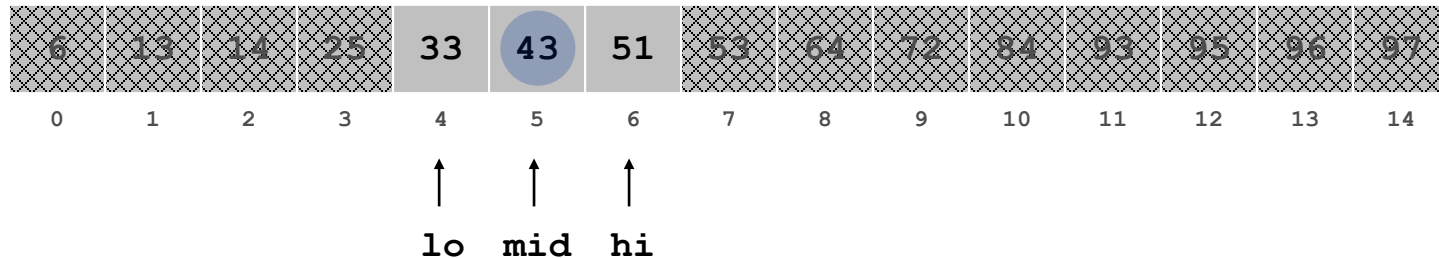
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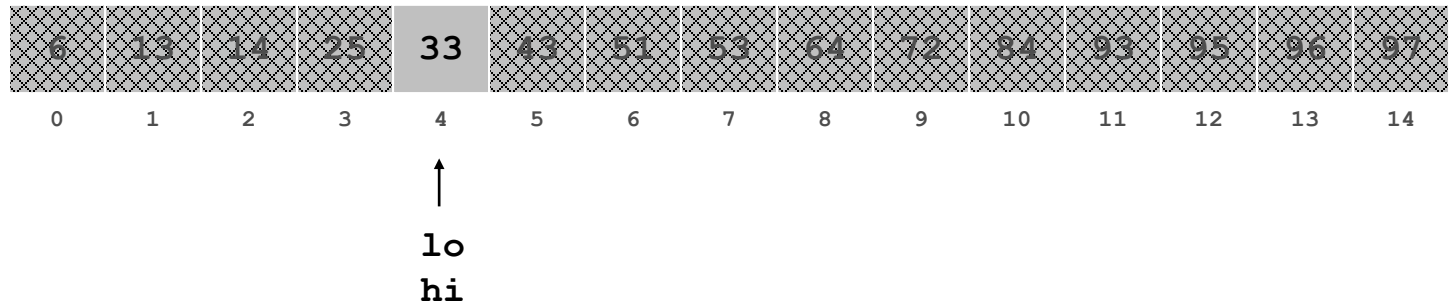


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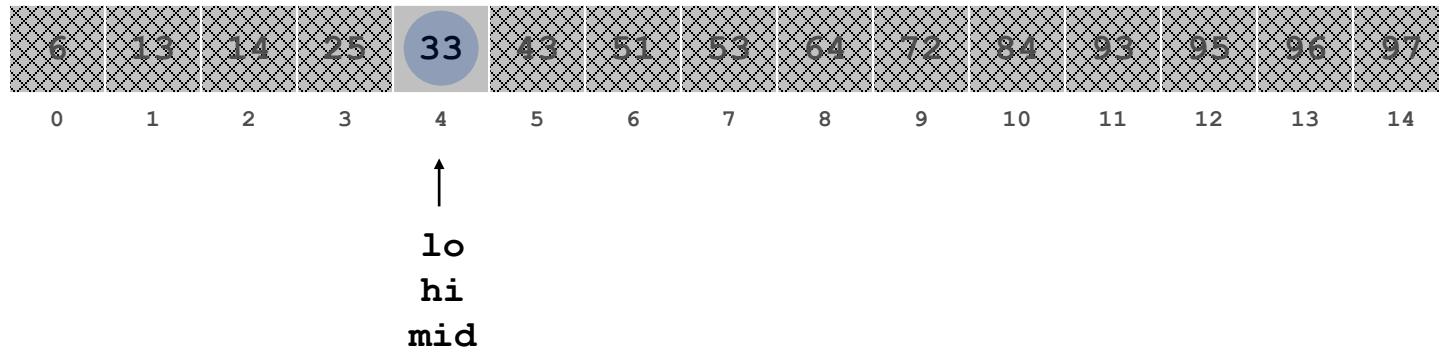


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